

MVD and Additional Coverage Checklist

Complete evidence map for Day 12 requirements and professional add-ons

Ruturaj Mokashi

Purpose

This checklist is the controlling evidence document. It maps the official MVD requirements and the additional agreed professional topics to the completed notebooks. The goal is to make missing coverage visible and prevent hidden gaps.

Coverage conclusion

All MVD requirements are covered. The additional topics identified during review are covered in Notebook 5 or final documentation. No item remains intentionally uncovered.

MVD coverage matrix

Area	Requirement	Evidence	Status
Beginner	Load Advertising and create TV vs Sales regression line	Notebook 1: regplot section and executed chart	Covered
Beginner	Train TV-only regression and print coefficient, intercept, MSE and R2	Notebook 1: simple model metrics	Covered
Beginner	Train multiple regression using TV, Radio and Newspaper; compare R2 and feature contribution	Notebook 1: comparison and contribution sections	Covered
Intermediate	Train Ridge, Lasso and ElasticNet with alpha values 0.1, 1, 10 and 100	Notebook 2: model loops and result tables	Covered
Intermediate	Plot RMSE against alpha for all three methods	Notebook 2: RMSE against alpha chart	Covered
Intermediate	Visualize Lasso coefficient path and identify features set to zero first	Notebook 2: Lasso path and zero feature table	Covered
Intermediate	Run LassoCV cv=5 and compare with unregularized model	Notebook 2: LassoCV and final comparison table	Covered
Expert	Train linear regression on Bike Sharing cnt and check residual normality with QQ plot	Notebook 3: linear baseline, residuals and QQ plot	Covered
Expert	Train PoissonRegressor and compare MAE and Mean Poisson Deviance	Notebook 3: PoissonRegressor comparison	Covered
Expert	Implement Negative Binomial Regression with statsmodels and explain when it is better than Poisson	Notebook 3: NB GLM, overdispersion and model decision	Covered
Topic title add-on	Cover Logistic Regression because the day title includes Linear and Logistic Regression	Notebook 4: full classification workflow	Covered

Additional topic coverage matrix

Topic	Evidence	Status
OLS vs Linear Regression	Notebook 1 and Notebook 5	Covered
Gradient descent vs closed-form OLS	Notebook 5	Covered
Correlation vs regression	Notebook 1	Covered
Interpolation vs extrapolation	Notebook 5	Covered
Residuals and error interpretation	Notebooks 1, 3 and 5	Covered

Topic	Evidence	Status
Regression assumptions	Notebooks 1, 3 and 5	Covered
Homoscedasticity and heteroscedasticity	Notebook 5	Covered
Outliers and influence	Notebook 5	Covered
Cook distance and leverage	Notebook 5	Covered
Confidence intervals and prediction intervals	Notebooks 1 and 5	Covered
p-values and statistical inference	Notebook 1 and Notebook 5	Covered
sklearn vs statsmodels difference	Notebooks 1, 3 and 5	Covered
Data leakage warnings	Notebooks 1 to 5	Covered
Business model recommendation table	Notebooks 2, 3, 4 and summary docs	Covered

Notebook 5 missing-topic closure

Notebook 5 was created after checking the coverage matrix. It closes the missing topics without disturbing the already approved notebook structure. This keeps the learning flow clean: core MVD tasks are in Notebooks 1 to 4, while deeper diagnostics and advanced add-ons are concentrated in Notebook 5.

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.